J1
Waveshare ESP32-S3-Zero
IO43 1
IO44 2
IO13 3
IO12 4
IO11 5
IO10 6
IO9 7
IO8 8
IO7 9
IO43
IO44
IO13
IO12
IO11
IO10
IO9
IO8
IO7
10 IO6
11 IO5
12 IO4
13
14 IO2
15 IO1
16 3V3
17 GND
18
IO6
IO5
IO4
IO2
IO1
3V3
GND
VBUS

DS1
CL3641AH common cathode
SEG_E 1
SEG_D 2
SEG_DP 3
SEG_C 4
SEG_G 5
DIG4 6
E
D
DP
C
G
DIG4
B
DIG3
DIG2
F
A
DIG1
7 SEG_B
8 DIG3
9 DIG2
10 SEG_F
11 SEG_A
12 DIG1

U1
SN74HC595N 3V3
DRV_B 1
DRV_C 2
DRV_D 3
DRV_E 4
DRV_F 5
DRV_G 6
DRV_DP 7
GND 8
QB
QC
QD
QE
QF
QG
QH
GND
QH_SERIAL
SRCLR_N
SRCLK
RCLK
OE_N
SER
QA
VCC
9
10 3V3
11 IO1
12 IO2
13 GND
14 IO43
15 DRV_A
16 3V3

R1
1k
DRV_A 1
1
2
2 SEG_A

R2
1k
DRV_B 1
1
2
2 SEG_B

R3
1k
DRV_C 1
1
2
2 SEG_C

R4
1k
DRV_D 1
1
2
2 SEG_D

R5
1k
DRV_E 1
1
2
2 SEG_E

R6
1k
DRV_F 1
1
2
2 SEG_F

R7
1k
DRV_G 1
1
2
2 SEG_G

R8
1k
DRV_DP 1
1
2
2 SEG_DP

R9
1k
IO4 1
1
2
2 BASE1

Q1
2N3904
GND 1
BASE1 2
E
B
C
3 DIG1

R13
10k
BASE1 1
1
2
2 GND

R17
10k
3V3 1
1
2
2 IO8

R10
1k
IO5 1
1
2
2 BASE2

Q2
2N3904
GND 1
BASE2 2
E
B
C
3 DIG2

R14
10k
BASE2 1
1
2
2 GND

R18
10k
3V3 1
1
2
2 IO9

R11
1k
IO6 1
1
2
2 BASE3

Q3
2N3904
GND 1
BASE3 2
E
B
C
3 DIG3

R15
10k
BASE3 1
1
2
2 GND

R19
10k
3V3 1
1
2
2 IO10

R12
1k
IO7 1
1
2
2 BASE4

Q4
2N3904
GND 1
BASE4 2
E
B
C
3 DIG4

R16
10k
BASE4 1
1
2
2 GND

R20
10k
3V3 1
1
2
2 IO11

SW1
MINUS two-prong
IO8 1
1
2
2 GND

SW2
SELECT two-prong
IO9 1
1
2
2 GND

SW3
PLUS two-prong
IO10 1
1
2
2 GND

SW4
RESOURCE two-prong
IO11 1
1
2
2 GND

D1
RED
GND 1
K
A
2 LED1_A

R21
1k
IO12 1
1
2
2 LED1_A

D2
RED
GND 1
K
A
2 LED2_A

R22
1k
IO13 1
1
2
2 LED2_A

D3
RED
GND 1
K
A
2 LED3_A

R23
1k
IO44 1
1
2
2 LED3_A

C1
100n
3V3 1
1
2
2 GND

C2
10u 10V
3V3 1
+
–
2 GND

C3
100n
3V3 1
1
2
2 GND

H1
M3

H2
M3

H3
M3

H4
M3

Q: 1=E, 2=B, 3=C. DS1: common cathode. R: 1/4 W.
J1: Waveshare ESP32-S3-Zero, 15.24 mm rows. U1: SN74HC595N at 3.3V.

Sheet: /
File: counter.kicad_sch

Title: Compact S3 / CL3641AH counter

Size: A2	Date:	**Rev: E1 prototype**
KiCad E.D.A. 10.0.4		Id: 1/1